BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/BEE 4750 /HW 2/hw2-regina`
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```
"""
                        (+)
Atmospheric CO          Global Mean Temperature

                                    (-)
                                  |
                            Ocean CO

         (-)         -
"""
```

```
"                        (+)\nAtmospheric CO          Global Mean Temperature\n
```

Problem 1 SOLUTION:

Please find a more detailed figure, labeled Figure 1, at the bottom of this PDF. This is incase the figure above is not showing correctly.

Henry's Law says that solubility of CO in seawater decreases as temperature increases. In other words, warm water holds less CO . Because of this, the arrow pointing from global mean temperature to the concentration of CO in the ocean is a negative (-) sign and this is a dampening relationship. As temperature increases, the ocean's water warms and CO concentrations decrease.

When concentrations of CO in the ocean increase, the concentration of CO in the atmosphere decrease since the ocean is acting as a carbon sink and taking out CO from the atmosphere. Because of this, the sign is negative (-) and this is a dampening relationship.

Finally, when CO concentrations in the atmosphere increases, the greenhouse effect is magnified since the CO is now absorbing heat and re-emitting it. As a result, the global mean temperature to increase. This is an amplifying relationship and the arrow's sign is positive (+).

This suggests that the overall feedback between temperature and the ocean carbon cycle is an amplyfing (or positive) feedback because an increase in atmospheric CO , increases temperature, which creates less CO2 in the ocean, which means less CO is being absorbed by the ocean and there is more of it in the atmosphere, which finally leads to even more warming.

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of $0.36\,\mathrm{d}^{-1}$ and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

2.1 Solution:

```
          Q1, C1                  Q2, C2
            |                       |
            |                       |
   -------|--------x=15 km   -----|-----
  |              |         |          |
Q---->           ----->Q3---->        ----> Q4
  |              |         |           |
      ---  --
```

Above is a rough idea of my systems diagram but please find a more detailed and clear version at the bottom of this PDF labeled as Figure 2.1.

Two boxes or control volumes are needed for this because there are two input streams, one starting at where the first wastewater input stream mixes into the river up until X = 15 km, right before the second wastewater stream mixes with the river. The second control volume captures the mixing of the second wastewater stream. These streams are causing a change in flow rate of the river so the control volumes will help model and understand those changes. Two equations are needed because the initial conditions are different for each - the river inflow before the first waste stream will have changed due to a few factors such as decay. This will have created a new initial condition of the water mixing with the second waste stream. This will be clearer below.

Problem 2.2

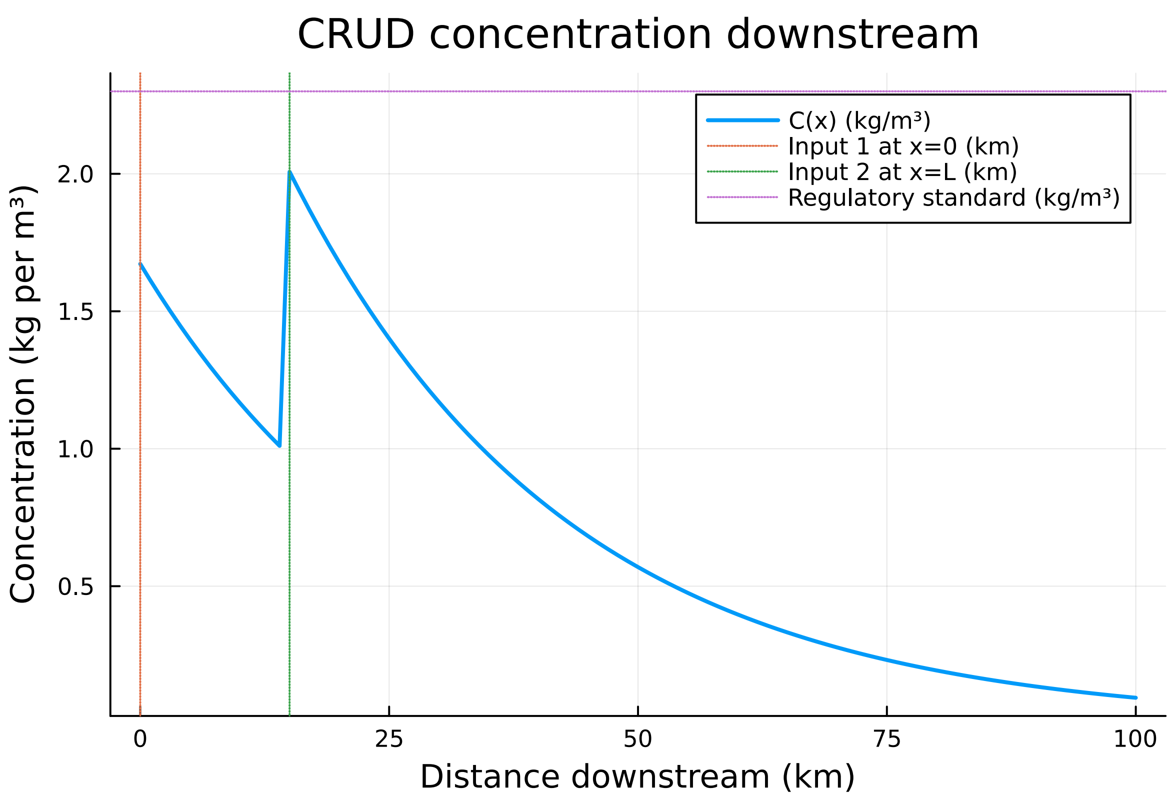
Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

```
Q = 250000 # river flow rate (m^3/d)
Ci = 0.5 # initial CRUD concentration in river (kg/m^3)
Q1 = 40000 # first wastewater input (kg/day)
C1 = 9 # concentration in first wastewater input (kg/m^3)   - SD   kg/1000m^3
Q2 = 60000 # second WW input (kg/m^3)       -SD should be m^3/day
C2 = 7 # concentration in second WW input (kg/1000m^3)

k = 0.36 # first order decay rate (1/day)
d = 54 # rate CRUD is deposited by the atmosphere along the river (kg/km/d)
v = 10 # constant average velocity of the river (km/day)
L = 15 # distance between WW input 1 and 2 (km)
A = Q / v


# fcn to find concentration at steady state
function steady_state_C(Q)
    # dC/dx = -decay rate + deposit rate = (-kC/v) + d/Q when dC/dx = 0 because steady state
    k = 0.36 # first order decay rate (1/day)
    d = 54 # rate CRUD is deposited by the atmosphere along the river (kg/km/d)
    v = 10 # constant average velocity of the river (km/day)
    d_c = d / Q   # (kg/km/d) ÷ (m³/d) = kg/km/m³   #     -SD added because d is kg/km/day,
    return((v*d_c)/(k*Q))  #     -SD changed d to c_d
end

# fcn to find the flow and concentration output of two streams mixing
```

```
function two_streams_mix(Q1, C1, Q2, C2)
    Q_out = Q1 + Q2
    C_out = ((Q1*C1) + (Q2*C2)) / Q_out
    return (Q_out, C_out)
end

# function conc_model(x, C0, Q)
#     k = 0.36 # first order decay rate (1/day)
#     d = 54 # rate CRUD is deposited by the atmosphere along the river (kg/km/d)
#     v = 10 # constant average velocity of the river (km/day)
#     C_out = []

#     C_out = steady_state_C(Q) .+ (C0 - steady_state_C(Q)) .* exp.((-k/v) .* (x))
#     return(C_out)
# end


# FIRST MIX: x = 0
(Q3, C3) = two_streams_mix(Q, Ci, Q1, C1)
println("Concentration at x = 0 km, right after first wastewater input: ", round(C3, digits=3


# conc at x = L- in reach 1
C_L = steady_state_C(Q3) + (C3 - steady_state_C(Q3)) * exp.((-k/v) * (15))
println("Concentration right at x = 15 km, right before second wastewater input: ", round(C_L

# SECOND MIX: x = 15 km
(Q4, C4) = two_streams_mix(Q3, C_L, Q2, C2)
println("Concentration right at x = 15 km, right after second wastewater input: ", round(C4,


function model(x)
    if x < L #before 15 km
        C_out = steady_state_C(Q3) .+ (C3 - steady_state_C(Q3)) .* exp.((-k/v) .* (x))
        C = C_out

    else #beyond 15 km
        C_out = steady_state_C(Q4) .+ (C4 - steady_state_C(Q4)) .* exp.((-k/v) .* (x-L))
        C = C_out
    end
    return C
end
```

```
c_down_stream = steady_state_C(Q4)
println("Steady state downstream: ", round(c_down_stream, digits=3),  " kg/m³")

x = 0:1:100
conc = model.(x)


p = plot(x, conc, lw=2, label="C(x) (kg/m³)",
         xlabel="Distance downstream (km)",
         ylabel="Concentration (kg per m³)",
         title="CRUD concentration downstream")
vline!(p, [0], ls=:dot, label="Input 1 at x=0 (km)")
vline!(p, [L], ls=:dot, label="Input 2 at x=L (km)")
hline!(p, [2.3], ls=:dot, label="Regulatory standard (kg/m³)")

display(p)
```




```
Concentration at x = 0 km, right after first wastewater input: 1.672 kg/m³
```

```
Concentration right at x = 15 km, right before second wastewater input: 0.975 kg/m³
Concentration right at x = 15 km, right after second wastewater input: 2.008 kg/m³
Steady state downstream: 0.0 kg/m³
```

I modeled the river by splitting it into two plug-flows with instantaneous complete mixing at each wastewater input with first-order decay and deposition occurring as well. At each input, I used point-mixing to get the post-mix concentration,

$$Q_{\text{out}} = Q_a + q_b, \qquad C_{\text{out}} = \frac{Q_a, C_a + q_b, c_b}{Q_{\text{out}}}.$$

Between inputs (and downstream of the second), concentration changes with distance x thanks to the first-order decay and deposition rate as stated above. The first-order rate is removing CRUD concentration and the deposition rate is adding concentration. I modeled this change in concentration using this equation:

$$\frac{dC}{dx} = -\frac{k}{v}C + \frac{s}{Q},$$

From that equation, I was able to manipulate it to reach the steady state equilibrium equation of

$$C_{\text{ss}}(Q) = \frac{v}{k}\frac{s}{Q^2},$$

and general solution of

$$C(x) = C_{\text{ss}}(Q) + (C_{\text{in}} - C_{\text{ss}}(Q))e^{-(k/v)x}.$$

Using the given flows and concentrations (all in kg/m³), I broke the model down into two equations:

$$C_1(x) = C_{\text{ss}}(Q_3) + (C(0^+) - C_{\text{ss}}(Q_3))e^{-(k/v)x}, \quad 0 \leq x \leq L.$$

and

$$C_2(x) = C_{\text{ss}}(Q_4) + (C(L^+) - C_{\text{ss}}(Q_4))e^{-(k/v)(x-L)}, \quad \text{for} x \geq L.$$

With these equations I got that the following concentration values at a few key distances:

$C(x = 0km) = 1.672kg/m^3$ - right after first wastewater input

$C(x = L^- = 15km) = 0.975kg/m^3$ - just before second wastewater input$$

$C(x = L^+ = 15km) = 2.008kg/m^3$ - immediately after second wastewater input

$C\text{ss}(Q4) = 0.0kg/m^3$ - further downstream at steady state

I plotted the concentration of CRUD as a function of distance downstream. After the first wastewater stream input until $x = 15$ km, the concentration decreases exponentially as first-order decay dominates; the uniform atmospheric deposition adds a small source of CRUD concentration, but not enough to offset the decay. At $x = 15$ km, the second discharge mixes in. Because this inflow adds more CRUD concentration to the river, the concentration sharply increases to $2.008 kg/m^3$. Farther downstream, the decay once again decreases the concentration and it trends toward a new steady-state value of $0.0 kg/m^3$ since there is no more wastewater inputs adding more CRUD.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Response

By observing the values of $C(0+)$, $C(L-)$, and $C(L+)$, along with the CRUD concentration graph, we can tell **the system is in compliance with the regulatory standard** of 2.3 kg/(1000m^3). In fact, the graph clearly shows that the regulatory standard is never exceeded! The stream after waste input 1 has a concentration of 1.672 kg/(1000m^3), which is within the regulatory standards and after traveling downstream the waste concentration in the stream continues to decrease exponentially. One the second source of waste enters the combined stream but only to 2.008 kg/(1000m^3), which is still below the regulatory standard of 2.3 kg/(1000m^3). After this second input, the stream once again decreases in CRUD concentration.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in $^\circ$C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/$^\circ$C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproteorozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);

- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Remove the greenhouse term, since it is not relevant in pre-industrial applications:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}}$$

Divide by C to isolate $\frac{dT}{dt}$ and distribute the negative:

$$\frac{dT}{dt} = \frac{1}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T\right) \tag{1}$$

Discretize the formula using the forward Euler method of the time derivative at t:

$$\frac{T(t+\Delta t) - T(t)}{\Delta t} = \frac{1}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T(t)\right) \tag{2}$$

Multiply by Δt and add $T(t)$ to the RHS to simplify the discretized differential equation:

$$T(t+\Delta t) = T(t) + \frac{\Delta t}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T(t)\right) \tag{3}$$

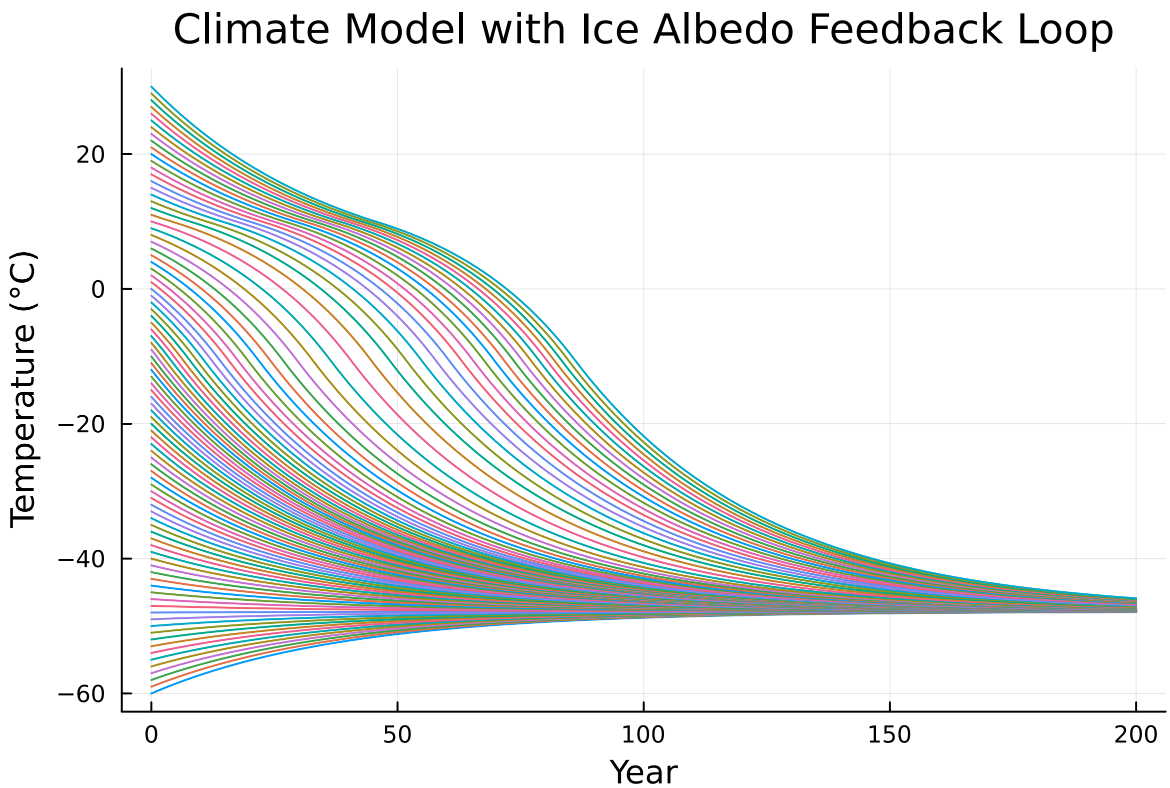
Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^{\circ}\text{C} \leq T_0 \leq 30^{\circ}\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
C = 51 # J/m^2/K
S = 1272 # W/m^2
A = 221.2 # W/m^2
B = -1.3 # W/m^2/C
ai = 0.5
ao = 0.3
dt = 1 # time step, yrs
n = 200 # yrs
To = -60:30 # range of initial temperatures, C

#Define an albedo function conditional on temperature
function a(T)
    if T <= -10
        return ai
    elseif T >= 10
        return ao
    else
        return ai + ((ao - ai) * ((T + 10)/20))
    end
end

#Define the simulation function of our discretized differential equation
#including all parameters and calling the albedo function defined above.
function albedo_feedback_sim(C, S, A, B, dt, n, T0)
    T = zeros(n + 1)
    T[1] = T0
    for i in 1:n
        T[i+1] = T[i] + (dt/C) * ((S * (1 - a(T[i])))/4 - A + (B * T[i]))
    end
    return T
end
#Simulate the model for all initial temperatures on [-60, 30]
sims = (T0 -> albedo_feedback_sim(C, S, A, B, dt, n, T0)).(To)
Ts = [t[1:n+1] for t in sims]
yrs = 0:n

#Plot graph of temperature over 200 years for all initial temperature conditions.
plot(yrs, Ts; legend=false, xlabel="Year", ylabel="Temperature (°C)", title="Climate Model wi
```



Based on the above graph and code, we can see that for all intial temperatures in the range $-60^{\circ}C$ to $30^{\circ}C$, the ice-albedo feedback hypothesis is does not cause significant enough warming to melt the "Snowball Earth" and reach a pre-industrial temperature of $14^{\circ}C$. For all intital temperatures, $-60^{\circ}C$ to $30^{\circ}C$ we see a single point of stability which occurs at $-46^{\circ}C$ for all intial temperatures.

In order to determine this, we used the simplified climate model from Question 3.1, which neglects the greenhouse gas term:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}}$$

We then discretized the differential equation using the forward (explicit) Euler method:

$$T(t+\Delta t) \;=\; T(t) + \frac{\Delta t}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T(t)\right) \tag{4}$$

From here, we must create a callable function that defines the albedo as a function of temperature according to the equation:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^{\circ}\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^{\circ}\text{C} \leq T \leq 10^{\circ}\text{C} \\ \alpha_0 & \text{if } T \geq 10^{\circ}\text{C}. \end{cases}$$

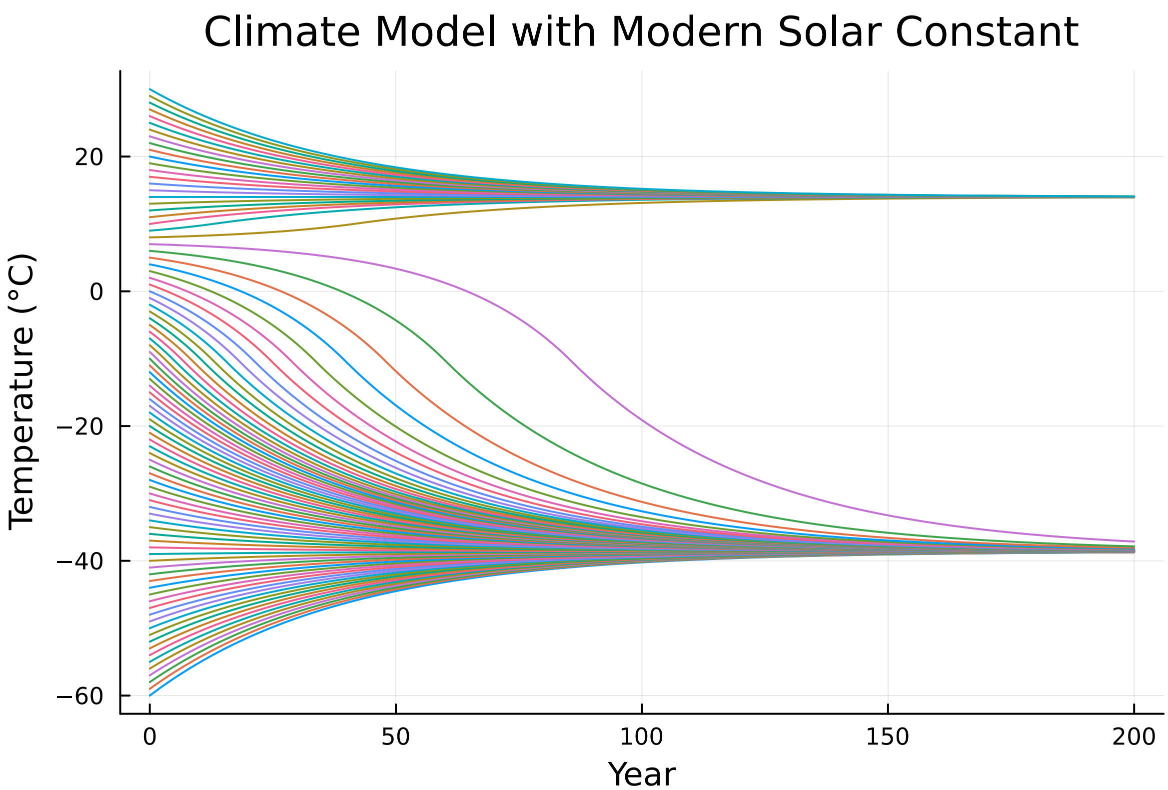
We can then create and run a simulation model for the discretized function with the intialized parameters, and call the albedo function for the varying intial temperatures over 200 years. This creates the model which shows a single equilibrium at $-46^{\circ}C$.

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

```
C = 51 # J/m2/K
S = 1368 # W/m2
A = 221.2 # W/m2
B = -1.3 # W/m2/C
ai = 0.5
ao = 0.3
dt = 1 # time step, yrs
n = 200 # yrs
To = -60:30 # range of initial temperatures, C

#Define an albedo function conditional on temperature
function a(T)
    if T <= -10
        return ai
    elseif T >= 10
        return ao
    else
        return ai + ((ao - ai) * ((T + 10)/20))
    end
end

#Define the simulation function of our discretized differential equation
#including all parameters and calling the albedo function defined above.
function albedo_feedback_sim(C, S, A, B, dt, n, T0)
```

```
    T = zeros(n + 1)
    T[1] = T0
    for i in 1:n
        T[i+1] = T[i] + (dt/C) * ((S * (1 - a(T[i])))/4 - A + (B * T[i]))
    end
    return T
end
#Simulate the model for all initial temperatures on [-60, 30]
sims = (T0 -> albedo_feedback_sim(C, S, A, B, dt, n, T0)).(To)
Ts = [t[1:n+1] for t in sims]
yrs = 0:n

#Plot graph of temperature over 200 years for all initial temperature conditions.
plot(yrs, Ts; legend=false, xlabel="Year", ylabel="Temperature (°C)", title= "Climate Model w
```



When looking at the Solar Constant, S, in the first model from Question 3.2, we used the value $S = 1272$ W/m^2 as this was the value of S in the Neoproterozoic Era. This resulted in a singular equilibrium at a value of $-46^{\circ}C$. When using the modern Solar Constant of $S = 1368$ W/m^2 for the same parameters and albedo function, we see 2 equilibria: for $8^{\circ}C$ to

$30^{\circ}C$, $14^{\circ}C$, and for $7^{\circ}C$ to $-60^{\circ}C$, $-46^{\circ}C$ - the pre-industrial temperature. This indicates that a solar constant of $S = 1368$ W/m^2, when combined with our specific assumptions and albeo model, is sufficient to cause warming to $14^{\circ}C$ for initial temperatures between $8^{\circ}C$ and $30^{\circ}C$.

References

List any external references consulted, including classmates.